

Pénélope Azuelos

email: penelope.azuelos@bristol.ac.uk
website: <https://penelopeazuelos.com>
address: Fry Building, School of Mathematics,
Woodland Rd, Bristol BS8 1UG, UK

Education

2022 – present	University of Bristol, UK <i>PhD in Mathematics (ongoing), supervised by Mark Hagen and Indira Chatterji</i>
2021 – 2022	École Normale Supérieure de Lyon, France <i>Master (M2) Mathématiques Avancées</i>
2017 – 2021	University of Bristol, UK <i>Master in Science in Mathematics (with first class honours)</i>

Publications

(with Damien Gaboriau) *Perfect kernel and dynamics: from Bass-Serre theory to hyperbolic groups*, *Mathematische Annalen* 391: 4733–4789, 2025.

<https://doi.org/10.1007/s00208-024-03038-w>

On subgroups with narrow Schreier graphs, *Bulletin of the London Mathematical Society*, 56(12): 3652–3668, 2024. <https://doi.org/10.1112/blms.13157>

Preprints

Periodic quasiflats in hierarchically hyperbolic spaces, arXiv:2608.01513, 2026.

Uncountably many homogeneous real trees with the same valence, arXiv:2511.03722.

A guide to constructing free transitive actions on median spaces, to appear in *Advances in Mathematics*, arXiv:2507.22230.

Teaching

Teaching assistant at the University of Bristol for the following courses:

Spring 2024	Introduction to proofs and group theory MATH10010
Autumn 2023	Combinatorics MATH30030
Spring 2020	Introduction to geometry MATH20004

Conference Talks

June 2026	Groups acting on non-positively curved spaces, International Academic Forum, Heidelberg, Germany
–	Branch groups: subgroups, rigidity, topologies, Universidad Complutense de Madrid, Spain
May 2026	Geometric group theory around cube complexes, CIRM, France
April 2026	Coarse geometry of groups, Centre Paul-Langevin, Aussois, France

February 2026	Median Geometry and Applications, MFO, Oberwolfach, Germany.
–	Orbit meetings, University of Warwick, UK
November 2025	Parole aux jeunes scientifiques du RT tétraèdre de Platon, Montpellier, France.
–	Operator algebras in the south of England, University of Bristol, UK.
June 2025	Conference on hyperbolicity and generalizations, Harbin Institute of Technology, China.
January 2025	RSME's 7th congress of young researchers, parallel session on geometric group theory, UPV/EHU, Bilbao, Spain.
September 2024	World of Geometric Group Theory, online.
March 2024	Rigidité en théorie des groupes : aspects géométriques et ergodiques, Université Paris-Saclay, Orsay, France.
May 2023	Measured Group Theory, Stochastic Processes on Groups and Borel Combinatorics, CIRM, Marseille, France.

Seminar Talks

May 2026	Geometric group theory seminar, University of Oxford, UK
April 2026	Junior geometry seminar, University of Bristol, UK
March 2026	Junior topology and group theory seminar, University of Oxford, UK
February 2026	DAGGER seminar, University of Warwick, UK.
–	Junior Geometry Colloquium, KIT, Karlsruhe, Germany.
January 2026	Geometry and Topology Seminar, University of Bristol, UK.
October 2025	Junior Algebra and Combinatorics Seminar, University of Bristol, UK.
September 2025	Séminaire Groupes et Actions, Université Paris-Saclay, Orsay, France.
February 2025	Junior Geometry Seminar, University of Bristol, UK.
December 2024	Séminaire d'Algèbres d'Opérateurs, IMJ-PRG, Paris, France.
October 2024	Séminaire de l'équipe d'algèbre, topologie et géométrie, Laboratoire J.A. Dieudonné, Nice, France.
September 2024	Séminaire de Théorie Ergodique, Institut de recherche mathématique de Rennes, France.
–	Algebra Geometry and Topology seminar, Heriot-Watt University, Edinburgh, UK.
May 2024	International Young Seminar on Bounded Cohomology and Simplicial Volume, online.
March 2024	GdT Action ! ENS de Lyon, France.
February 2024	Junior Geometry Seminar, University of Cambridge, UK.
December 2023	Postgraduate Seminar, University of Bristol, UK.
November 2023	Junior Topology and Group Theory Seminar, University of Oxford, UK.
September 2023	Le séminaire virtuel francophone Groupes et Géométrie, online.
January 2023	Junior Geometry Seminar, University of Bristol, UK.

Service

2025	Organisation of the Postgraduate Group Theory Conference 2025
2023 – 2024	Organisation of the Bristol Junior Geometry Seminar

Outreach

July 2026	Talk and panel participation at an undergraduate summer school for women in mathematics, University of Warwick
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Awards and funding

2024	Grant from the Académie d'Excellence "Systèmes Complexes" for a three month research visit to the LJAD in Nice, supervised by Indira Chatterji.
2022 – 2026	University of Bristol PhD scholarship.
2021 – 2022	MiLyon scholarship for an M2 masters program.
2020	Summer research bursary funded by the Heilbronn Institute for Mathematical Research, supervised by Misha Rudnev.

Languages

English and French (native speaker)